An Analysis of How Learning Methods Impact Academic Performance

CUS 1179: Machine Learning

By: Avneet Kaur, Geovani Tzul, Hawa Drame, Jericho McQueen, Javokhir Akhmadjonov

Dr. Parisi

1.0- Abstract

This study uses statistical analysis and machine learning approaches to investigate how student actions and learning strategies affect academic performance. The analysis concentrated on whether student exam scores were impacted by learning style, study time, attendance, and smartphone use. The findings indicated that although the differences were negligible, students in mixed and offline learning contexts performed somewhat better than those in online settings. Smartphone use had a weak negative relationship with exam scores, but study hours had the strongest positive correlation. Predictive performance was assessed using Random Forest Regression, Decision Tree Regression, and Multiple Linear Regression; however, the explanatory power of each model was limited. Overall, the results imply that learning strategies and student behavior alone are insufficient to adequately explain academic achievement, which is impacted by a variety of interrelated factors.

2.0 - Introduction

In recent years, education has changed a lot, especially with the rise of online learning and technology use among students (Alarifi & Song, 2024). Learning is no longer limited to traditional classrooms, and students now have options such as online, offline, and mixed learning environments (Alarifi & Song, 2024). Because of this shift, it has become important to understand how these different learning methods affect student academic performance (Kandukoori et al., 2024). At the same time, student behaviors such as study habits, attendance, and smartphone usage may also play a role in shaping outcomes (Hannay and Newvine, 2006).

This study focuses on analyzing how different learning methods impact student exam performance. It looks at whether students perform differently in online, offline, or mixed learning environments. It also examines how study behaviors, including study hours and smartphone usage, influence exam scores. The main research question is: *Do learning methods and student behaviors significantly affect academic performance?*

To answer this question, this paper uses a dataset of high school students and applies both exploratory data analysis and predictive modeling techniques. The analysis explores patterns and relationships between variables and then uses a regression model to evaluate how well these factors can predict exam scores. Overall, the goal is to better understand the factors that contribute to student success and identify whether the learning method plays a meaningful role.

3.0- Literature Review

Recent research on higher education instruction increasingly focuses on the role of digital learning technologies and how they compare with traditional face-to-face teaching methods. The articles reviewed by our team highlight three recurring themes in the literature: (1) student-centered digital tools can enhance engagement and motivation, (2) differences between online and in-person instruction depend heavily on course design and student characteristics, and (3) engagement and access to learning resources often mediate academic outcomes.

One important contribution to this discussion is the work of Calderón, Meroño, and MacPhail (2020), summarized by Javokhir in the group research notes. Their study examined a student-centered digital learning approach that incorporated blogs, multimedia assignments, and collaborative online discussions. The researchers found that when digital tools were integrated into coursework through interactive and project-based activities, students reported higher intrinsic motivation and improved perceptions of the classroom climate. Additionally, these motivational improvements were associated with measurable increases in exam performance. This research suggests that the effectiveness of digital learning technologies is strongly tied to the way they are implemented. Simply adding technology to a course may not improve outcomes unless the tools encourage active participation and collaborative learning.

Another perspective is provided by Alarifi and Song (2024), whose study compares online and in-person learning outcomes in higher education. As summarized by Hawa, the authors found that student performance sometimes favors traditional classroom environments, particularly for students early in their academic careers who may require more structured learning environments. However, the study also shows that well-designed online courses can achieve similar or even better results than traditional courses when they incorporate strong instructional design and interactive components. These findings highlight the importance of considering contextual factors such as course structure, instructor support, and student self-regulation when evaluating the effectiveness of digital learning environments.

Further evidence comparing digital tools and traditional teaching methods is provided by Kandukoori, Kandukoori, and Wajid (2024). Avneet's summary of this study indicates that students using interactive digital resources demonstrated significantly larger improvements in test scores between pre- and post-assessments compared with students relying primarily on traditional worksheets and textbooks. In the reported results, the digital-learning group achieved approximately a 24 percent improvement in scores, while the traditional group improved by roughly 8 percent. Although the findings are promising, the authors acknowledge that sample size and implementation conditions may influence the magnitude of these effects.

Finally, Hannay and Newvine (2006) explore student perceptions of distance learning in comparison to traditional classroom instruction. Geovani's notes on this article highlight that students enrolled in online courses often report spending more time studying and engaging with course materials. However, the authors also note that many students choose online courses because of scheduling flexibility or work commitments. This self-selection effect means that differences in outcomes between online and traditional learning environments may partly reflect differences in the students who choose those formats rather than the instructional method itself.

Taken together, these studies demonstrate that digital learning tools have significant potential to enhance academic outcomes, particularly when they promote active learning and collaboration. However, the literature also emphasizes that technology alone does not guarantee improved performance. Student characteristics, engagement levels, and course design all influence whether digital or traditional instruction produces better results. These insights inform the analytical strategy for our project, which treats learning method as one factor among many influencing student success.

4.0 - Methodology

4.1- Overview of Methodology

This study examines how learning methods and student behaviors influence academic performance using the Student Academic Performance Dataset obtained from Kaggle. The dataset consists of high school students (Grades 9–12) and includes both academic and behavioral variables. Key features in the dataset include **study hours**, **attendance percentage**, **exam scores**, **subject**, **grade level, learning method** (online, offline, mixed), and **smartphone usage** (hours per day). These variables allow for both behavioral and instructional factors to be analyzed simultaneously.

4.2- Data preparation and cleaning

The dataset was first loaded and examined to understand its structure, variable types, and overall quality. Each observation represents an individual student and includes both numerical and categorical attributes. Study hours, attendance, and exam scores were treated as continuous variables, while subject and learning method were treated as categorical variables.

Data preprocessing was conducted to ensure the dataset was suitable for analysis. Categorical variables such as learning method, and subject were converted into numerical format using encoding techniques to allow their inclusion in statistical models. The dataset was also checked for missing values and inconsistencies; however, only minimal issues were identified, and no significant imputation was required. In addition, potential outliers in variables such as study hours and exam scores were examined to ensure that extreme values did not disproportionately influence the results. These preprocessing steps helped ensure that the dataset was clean, consistent, and appropriate for further analysis. Since the dataset utilized in this work is anonymised and publicly accessible, no personally identifying information is included, the data acquisition raises no direct ethical or privacy issues.

4.3- Exploratory Data Analysis

Exploratory data analysis was conducted to identify patterns, trends, and relationships among the variables. Summary statistics, including the mean, median, and standard deviation, were calculated for all numerical variables to provide an initial understanding of the data distribution. In addition, several visualizations were created to further explore relationships between key variables.

Bar charts were used to compare average exam scores across different learning methods, including online, offline, and mixed formats. Scatter plots were generated to examine the relationship between study hours and exam scores, as well as between smartphone usage and academic performance. A correlation matrix was also constructed to quantify the strength and direction of relationships among all numerical variables.

The results of the exploratory analysis indicated generally weak relationships between individual variables and exam scores. This suggests that academic performance is influenced by multiple interacting factors rather than a single dominant predictor.

4.4- Analytical Approach and Predictive Modeling

This study included both exploratory and predictive analytical methods to investigate the relationship between student actions and instructional elements and academic performance. To find general trends in the data, exploratory techniques such correlation analysis, scatterplots, and descriptive statistics were employed.

Several supervised learning models were used to assess predicted performance. Because of its interpretability and capacity to estimate straightforward linear connections between predictors and exam scores, a multiple linear regression model was initially employed as a baseline method.

More adaptable non-linear models were also tried because early exploratory results indicated weak linear correlations among multiple variables. In particular, Random Forest Regression and Decision Tree Regression were used to better capture non-linear patterns and interaction effects that linear regression could miss.

Root Mean Square Error (RMSE) and R2 were used to compare model performance. RMSE was used to measure prediction error, and R2 was used to determine the percentage of variation that each model explained. The study was able to ascertain whether more sophisticated machine learning techniques outperformed conventional linear regression in terms of predicted accuracy by contrasting these models.

4.5- Summary

Overall, this methodology integrates exploratory data analysis with predictive modeling to provide both descriptive and analytical insights into student academic performance. By incorporating behavioral and instructional variables, the approach captures the multifaceted nature of academic success while maintaining clarity and interpretability in the analysis.

5.0 - Analysis and Results

5.1- Overview of Analysis:

This study examines the relationship between learning environments, study habits, and technology use and student academic achievement using both exploratory data analysis and predictive modeling methodologies. Using descriptive statistics and visualizations, the investigation first investigates broad trends in exam scores across various learning contexts and student behaviors. Next, it assesses the predictive power of both conventional statistical modeling and more sophisticated machine learning techniques for academic success. This part evaluates if there are variations between student groups as well as how well academic outcomes can be explained and predicted using the given variables by combining exploratory analysis with model comparison.

5.2- Learning Method and Academic Performance

A comparison of average exam scores across learning methods was conducted:

Figure 1: Average Exam Score by Learning Method

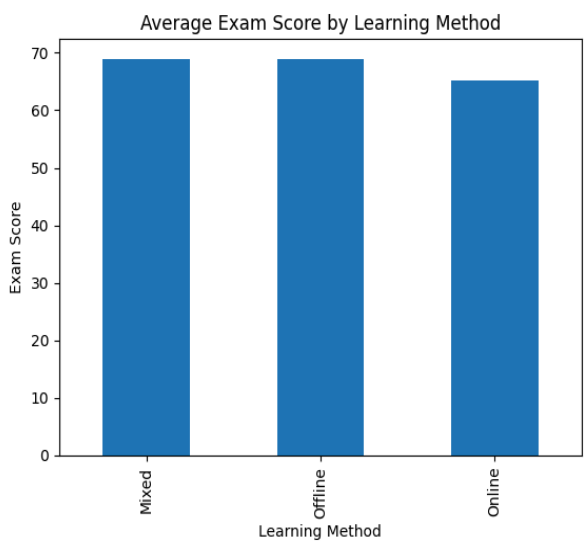
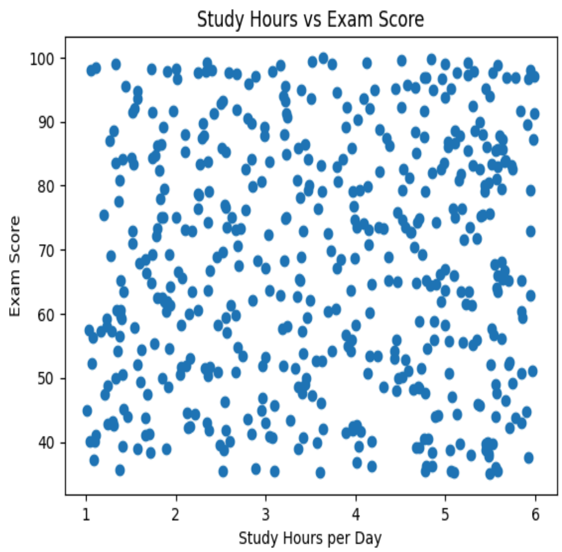


Figure 1 illustrates that the average exam scores of students in Mixed Learning and Offline environments were marginally higher than those of students in online learning methods. The comparatively small group differences, however, indicate that variance in academic achievement cannot be well explained by learning style alone. Although there is some diversity among instructional forms, the findings suggest that behavioral and academic characteristics other than the learning environment probably have an impact on student achievement.

5.3- Study Hours and Exam Performance

The relationship between study hours and exam scores was examined:

Figure 2: Study hours vs Exam Score

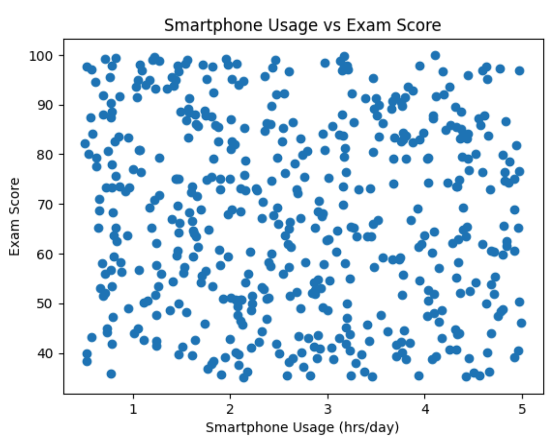


In Figure 2, exam scores vary significantly across all study time levels and there is no discernible linear relationship between study hours and academic achievement. Exam results are both high and low at almost every level of study time, indicating that study time is not a reliable indicator of academic success. The result suggests that other academic and behavioral factors probably affect exam performance, even though study habits may still have an impact on student outcomes (Calderón et al., 2020).

5.4- Smartphone Usage and Exam Performance

The analysis also examined the relationship between smartphone usage and exam scores:

Figure 3: Smartphone Usage vs Exam Score



In Figure 3, there is no clear linear correlation between smartphone use and academic achievement which displays significant variation in exam scores across all smartphone usage levels. Nearly all smartphone usage levels show both high and low exam scores, indicating that smartphone use by itself is not a reliable independent predictor of exam performance. The data suggests that academic performance is probably impacted by behavioral and academic issues other than smartphone usage alone, even though excessive technology use may nevertheless occasionally contribute to distraction (Hannay & Newvine, 2006).

5.5- **Correlation Analysis**

A correlation matrix was generated to analyze the strength and direction of relationships between key variables and exam scores:

Figure 4: Correlation Matrix

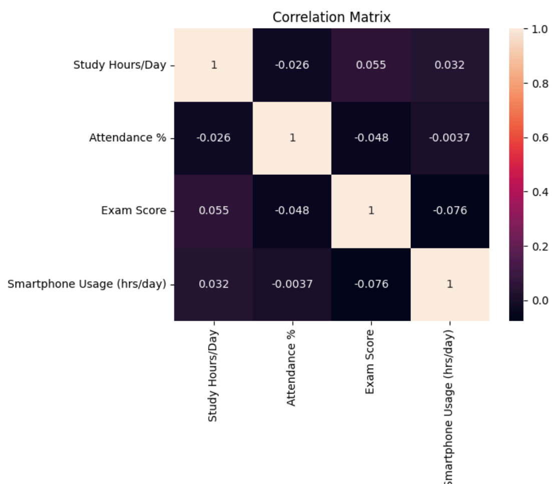


Figure 4 illustrates the associations between exam scores and the other factors are all extremely weak and nearly negligible. Exam scores are only slightly positively correlated with study hours, whereas attendance and smartphone use are negligibly negatively correlated. These findings imply that there isn't a single factor that strongly correlates with academic achievement. Because the variables in the model do not show significant independent linear connections with exam scores, this helps explain why the linear regression model demonstrated low predictive strength.

5.6- **Linear Regression Results**

Study hours, attendance, smartphone use, learning style, and grade level were used as variables in a multiple linear regression model to assess the predictive association between student attributes and exam scores.

Table 1:

Predictor	Coefficient (β)	p-value	Significant?

Study Hours/Day	1.565	0.016	Yes
Attendance %	-0.103	0.220	No
Smartphone Usage (hrs/day)	-0.571	0.432	No
Class/Grade_Grade 11	3.871	0.144	No
Class/Grade_Grade 12	4.489	0.082	No
Class/Grade_Grade 9	1.258	0.643	No
Learning Method_Offline	-0.577	0.806	No
Learning Method_Online	-3.676	0.112	No
Favorite Subject_English	-4.937	0.099	No
Favorite Subject_History	-9.164	0.003	Yes
Favorite Subject_Math	-4.922	0.090	No
Favorite Subject_Science	0.280	0.924	No

Model Fit Statistics: $R^2 = 0.067$, Adjusted $R^2 = 0.038$, RMSE = 18.63

Figure 5: Actual vs Predicted Exam Scores

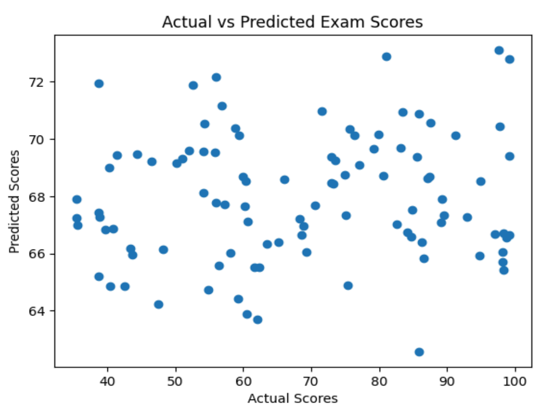


Figure 5 and Table 1 illustrates that exam scores were predicted using a multivariate linear regression model that took into account study hours, attendance, smartphone usage, grade level, preferred subject, and learning style. This enlarged regression incorporated categorical classroom features along with behavioral and academic predictors, in contrast to the previous basic model.

With an R2 of 0.067 and an adjusted R2 of 0.038, the model's predictors accounted for around 6.7% of the variation in exam scores. The model offers a more thorough and statistically sound evaluation than the previous single-variable method, despite the fact that this shows poor explanatory ability overall.

The most statistically significant variable among the predictors was study hours per day (β = 1.565, p = 0.016), suggesting that, after adjusting for other variables, more study time was linked to somewhat better exam performance. On the other hand, in the whole model, smartphone use, attendance, and learning style were not statistically significant independent predictors. While the majority of grade-level and subject-category variables were not significant, students who preferred history had considerably lower scores than the reference group (β = -9.164, p = 0.003).

The average difference between the anticipated and actual exam scores was almost 19 points, as indicated by the model's Root Mean Square Error (RMSE) of 18.63. This suggests that there is still a significant amount of unexplained variation in student performance even though the model captures certain basic patterns.

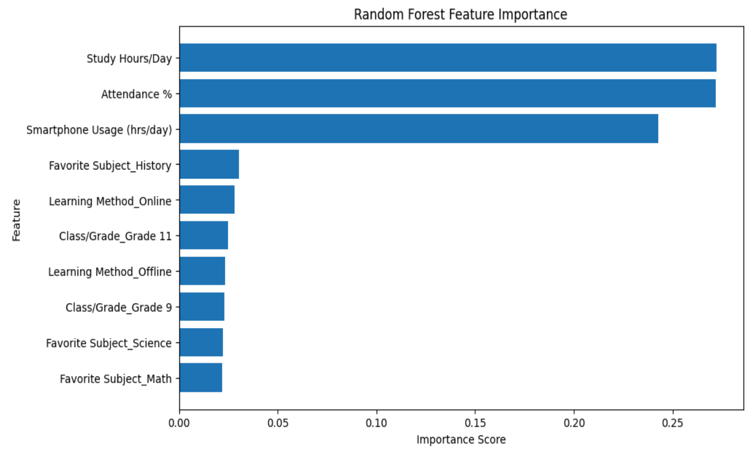
5.7 - Advanced Model Comparison (Random Forest Regression)

Using the same predictors as the expanded linear regression model, a Random Forest and Decision Tree Regressor was used to assess if a more advanced machine learning model could enhance prediction performance.

Table 2:

Model	RMSE	R^2
Multiple Linear Regression	18.63	0.067
Decision Tree Regression	20.15	-0.365
Random Forest Regression	18.03	-0.092

Figure 6: Random Forest Feature Importance



To explore if more adaptable nonlinear models could enhance prediction performance, Decision Tree Regression and Random Forest Regression were used with the same predictors as the expanded linear regression model (Figure 6 and Table 2). In order to determine whether nonlinear splits and interactions may more accurately represent exam score fluctuation without assuming linear relationships, Decision Tree Regression was initially employed. Nevertheless, the Decision Tree model performed worse than the linear regression baseline, yielding the weakest explanatory fit ($R^2 = -0.365$) and the highest prediction error (RMSE = 20.15), suggesting that a single nonlinear tree most likely overfitted the training set and performed poorly when applied to new observations.

The instability of a single decision tree was then lessened by using Random Forest Regression as a more stable ensemble-based nonlinear model. The Random Forest model's R2 value remained negative (-0.092), indicating that it was still unable to adequately explain exam score fluctuation, even though it somewhat increased prediction accuracy by reducing RMSE to 18.03 from 18.63 in linear regression. Overall, these results imply that the low explanatory strength of the available variables remained the fundamental constraint and that increasing model complexity alone did not significantly enhance predictive performance.

6.0 - Model Evaluation and Improvement

Root Mean Square Error (RMSE) and R2 were used to measure the prediction accuracy and explanatory power of the model. The expanded multiple linear regression model yielded an R2 of 0.067 and an RMSE of 18.63, suggesting that although the model identified some broad patterns, it only partially explained the variation in exam scores. Using the same predictors, a Random Forest Regressor was also used to see if a more sophisticated modeling technique could enhance performance. With an RMSE of 18.03, the Random Forest model marginally decreased prediction error, but its R^2 value of -0.092 demonstrated that it was still unable to adequately explain exam score variation. Random Forest outperformed linear regression in capturing nonlinear interactions, although the improvement was negligible, indicating that model complexity was not the main obstacle.

Both models' poor performance suggests that the explanatory power of the available predictors was limited. Even though factors including study time, attendance, smartphone use, and learning style had some predictive value, they were insufficient to effectively estimate academic achievement by themselves. This implies that characteristics like motivation, past academic performance, socioeconomic background, learning support, or instructional quality that are not included in the dataset probably have an impact on student results (Alarifi & Song, 2024). Future advancements would probably rely more on growing the dataset to incorporate stronger explanatory variables than on making the model more advanced (Kandukoori et al., 2024).

7.0 - Conclusion and Discussion

This study's primary goal was to ascertain whether student actions and learning strategies have a major impact on academic achievement. The results indicate that a variety of interrelated factors, including study habits, technology use, and instructional format, influence academic success more so than any one factor working alone (Alarifi & Song, 2024). Although the average exam scores of students in mixed and offline learning contexts were marginally higher than those in online settings, these differences were negligible, suggesting that learning format by itself was not a reliable indicator of success.

Study time was the variable that consistently exhibited a positive link with exam scores, but smartphone usage showed a weak negative relationship, indicating that academic performance may be slightly impacted by distraction (Hannay & Newvine, 2006). Regression and correlation analysis, however, revealed that the majority of individual predictors had poor explanatory power. Only a small percentage of exam score fluctuation was explained by the multiple linear regression model, and while Random Forest regression lowered prediction error somewhat, its overall predictive value was also restricted. This implies that the low explanatory power of the available variables, rather than model selection alone, was the main constraint.

Overall, the results show that a wider range of behavioral, contextual, and personal characteristics than those included in this dataset have an impact on academic success. Although they were not included, factors that probably have a significant impact on student results include motivation, past academic competence, socioeconomic background, and instructional quality. To better understand and estimate academic achievement, future research should take these other variables into account and investigate more thorough predictive frameworks.

References

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